BM CLOUDHIRE AI

Enterprise Recruitment SaaS

System Architecture & Technical Specification

A comprehensive guide to the architecture, automated candidate matching rubric, cloud infrastructure, and user workflows.

APPLICATION
BM CloudHire AI

HOSTING & CLOUD
AWS Amplify • Lambda • Bedrock

PRODUCTION STATUS
Live on AWS Amplify

1. Executive Summary

BM CloudHire AI is an enterprise-grade cloud recruitment platform engineered to automate and optimize candidate evaluation. The platform bridges the gap between hiring managers and applicants by leveraging **Amazon Bedrock Foundation Models** to perform multi-criteria technical resume parsing and candidate ranking on an objective **100-point rubric**.

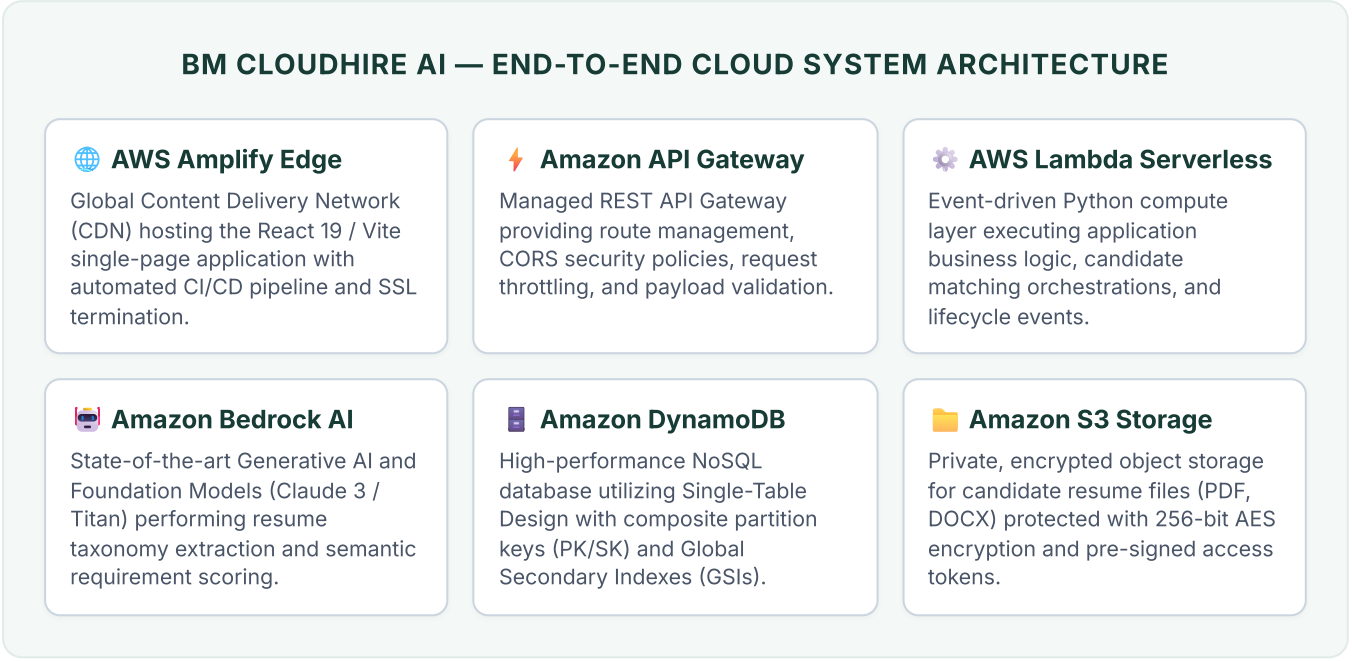
The platform eliminates manual resume screening delays, guarantees strict data privacy, and enforces standardized candidate evaluations across diverse engineering and cloud specializations.

Key Business Objectives

- **Accelerated Time-to-Hire:** Automated resume ingestion and AI matching evaluates candidate pools in seconds rather than days.
- **Objective Scoring:** Transparent 100-point scoring formula prevents unconscious recruiter bias.
- **Enterprise Security:** Resumes are securely stored in private Amazon S3 buckets with time-limited pre-signed access URLs.
- **Strict Identity Isolation:** Administrator sessions are strictly masked as *Administrator* (admin@bmcloudhire.com), protecting individual recruiter personal data.

2. Cloud System Architecture

The system is built on a serverless, highly available **AWS Cloud Architecture** designed for low latency, auto-scaling, and resilience.



3. The 100-Point Candidate Evaluation Rubric

The platform employs an objective, 5-tier technical scoring rubric powered by Amazon Bedrock. When a job requisition is matched against available candidates, each applicant is evaluated across the following transparent dimensions:

1. Required Mandatory Skills Match (Must-Have Technologies) — **40 Points (40%)**

Evaluates exact presence of mandatory skills (e.g., Python, AWS Lambda, DynamoDB, React). Each verified skill awards proportional score.

2. Hands-on Professional Experience (Years of Practice) — **25 Points (25%)**

Evaluates verified career seniority against the role's minimum threshold. Full marks awarded when verified experience meets or exceeds requirements.

3. Technical Project Deliverables & System Architecture — **20 Points (20%)**

Analyzes real-world deliverables, production deployment experience, microservices development, and telemetry monitoring in past projects.

4. Preferred & Bonus Skills (Infrastructure & DevOps) — **10 Points (10%)**

Awards bonus credit for secondary tooling such as Docker, Kubernetes, Terraform, AWS CDK, GraphQL, and CI/CD automation pipelines.

5. Education Standard & Industry Certifications — **5 Points (5%)**

Validates computer science or engineering degree credentials along with active AWS professional certifications.

Candidate Qualification Tiers

SCORE RANGE	CLASSIFICATION	BADGE COLOR	RECOMMENDED RECRUITER ACTION
80 – 100 Points	**Top Match (Tier 1)**	Mint Pale Badge	Immediate fast-track advancement to technical interview round.
60 – 79 Points	**Qualified (Tier 2)**	Green Badge	Shortlisted for secondary review or telephone screening.
Below 60 Points	**Needs Assessment (Tier 3)**	Amber Badge	Hold in repository for alternate requisitions or send info request.

4. User Personas & Core Workflows

Candidate Workflow

1. **Account Creation:** Candidate enters full name, email address, and secure password on the public portal.
2. **Job Exploration:** Browses open vacancies on the career board filtered by role title, technical domain, and location.
3. **One-Click Application:** Selects a vacancy, reviews required/preferred skills, and submits their application with resume attachment.
4. **Application Tracking:** Monitors real-time status in the *My Applications* center (*Submitted*, *Under Review*, *Approved*).

Administrator & Recruiter Workflow

1. **Secure Authentication:** Signs in to the enterprise workspace with administrative credentials. Profile is automatically locked to **Administrator**.
2. **Requisition Oversight:** Uses the **Requisition Picker Grid** to view applicant pools per job.
3. **Streamlined Candidate Review:** Evaluates applicants in the standardized table with single-line skill chips, verified experience, and AI scores.
4. **Decision Execution:** Advances top candidates with **Approve (Next Stage)**, triggering state transitions and updating metric counters.
5. **AI Matching Execution:** Launches the Bedrock Matching Engine to calculate rubric rankings and assign top candidates.
6. **Audit Ledger:** Inspects chronological event timestamps in *Assignments & History* for compliance.

5. Security, Identity & Enterprise Design Standards

Identity Masking & Data Privacy

- **Zero Personal Exposure:** To protect hiring leads and system operators, administrative accounts strictly render as **Administrator** with the **AD** avatar across all UI elements, top headers, dropdowns, and sidebars.
- **Pre-Signed S3 Access:** Candidate resumes cannot be directly scraped or browsed publicly. Download links expire automatically after 15 minutes.
- **Password Hashing:** Credentials utilize secure SHA-256 / bcrypt hashing with salting.

Enterprise Forest & Mint Design System

DESIGN ELEMENT	HEX CODE	VISUAL ROLE IN INTERFACE
Forest Green	`#173C35`	Primary sidebar background, primary submit buttons, brand logo marks.
Mint Green	`#52BFA0`	Active button hover states, focus rings, match badges, accent pills.
Pale Mint	`#DDF5EB`	Active navigation indicator tabs, selected requisition card fills.
Page Background	`#F4F8F6`	Clean, light-tinted canvas ensuring low eye strain during long sessions.
Card Container	`#FFFFFF`	Pure white cards with subtle 1px border (`#E2ECE7`) and soft drop shadows.
Text Charcoal	`#24352F`	High-contrast, readable typography complying with WCAG AAA accessibility.

6. Credentials & Execution Reference

PORTAL ROLE	USERNAME / EMAIL	PASSWORD	PLATFORM PRIVILEGES
Administrator	`admin` or `admin@bmcloudhire.com`	`boopathi_2004`	Full administrative control, job publishing, applicant review/approval, AI matching, system audit ledger, platform settings.
Candidate	`arun.krishnan@example.com`	`Candidate@2026`	Career search, 1-click application submission, real-time stage tracking, personal profile and resume upload.

Maintenance & Deployment Commands

ACTION	EXECUTION COMMAND LINE
Run Local Dev Server	`cd frontend && npm run dev`
Build Production Bundle	`cd frontend && npm run build`
Deploy to AWS Amplify	`python backend/scripts/deploy_to_amplify.py`
Automated QA Chrome Suite	`python backend/scripts/full_suite_test.py`
Generate System PDF	`python backend/scripts/generate_project_pdf.py`